

Did Generative AI Trigger a Structural Break? ARIMA Analysis of XLK ETF

1 Introduction

Artificial Intelligence (AI) has been a hot topic in recent years, with its wide applications in scientific discovery, advanced cybersecurity, and personalized AI agents. In the stock market, technology stocks have accounted for 32% of global equity returns and 40% of total U.S. equity market returns since 2010 (Goldman Sachs, 2024), playing a dominant role in modern financial markets. Large-cap

With the crucial role of technology stock in the general financial market and AI product's impacts on company's evaluation, we want to investigate whether the technological breakthrough impact the technology sector stock market. This report investigates:

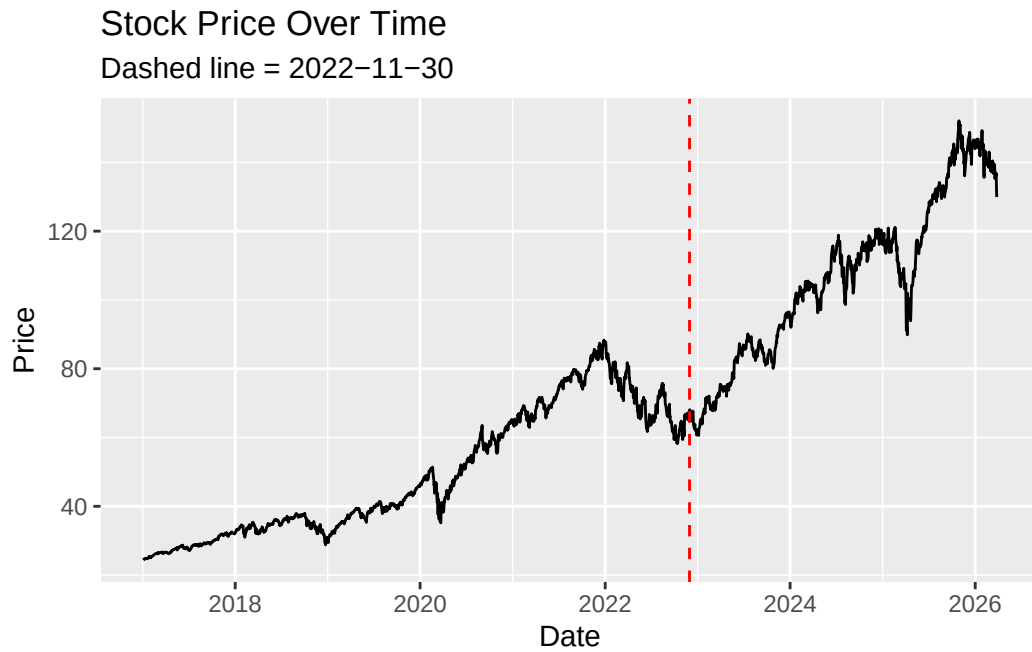
- Did the stock return pattern changes after the emergence of AI products?
- What's the predicted stock return based on current stock market trend?

Specifically, we evaluate this change by analyzing the Technology Select Sector SPDR Fund (XLK) using ARIMA models to examine whether the release of ChatGPT corresponds to a statistically significant shift in XLK price dynamics from 2017

2.2 Data Collection

Using the dataset from the investing website, we derive the following variables for the analysis:

- **Price** (in USD): The final price at which the ETF traded when the stock exchange closed (typically 4:00 PM EST).
- **Open** (in USD): The price of the very first transaction when the market opened (9:30 AM EST).
- **High & Low** (in USD): The maximum and minimum prices reached during the



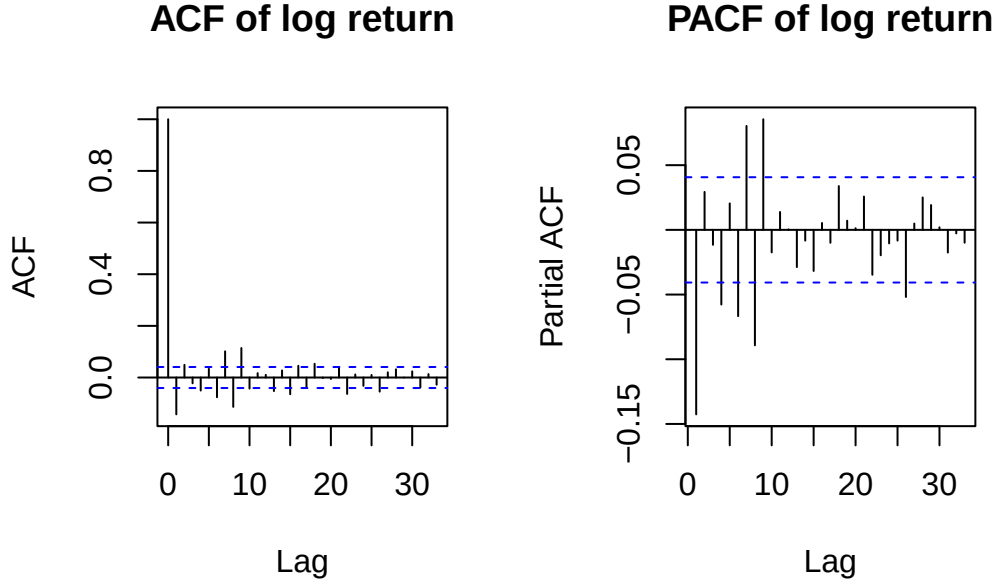
The autocorrelation function (ACF) plot in Figure 2 demonstrates a strong correlation between data points, indicating this time series is non-stationary and does not exhibit white noise characteristics. The ACF values persist well beyond the significance boundaries and decay at a very slow rate. This high correlation is explainable since the stock prices are accumulated based on the previous day's close price. Therefore, data transformation is required to achieve stationarity prior to modeling.

3 Methods and Results

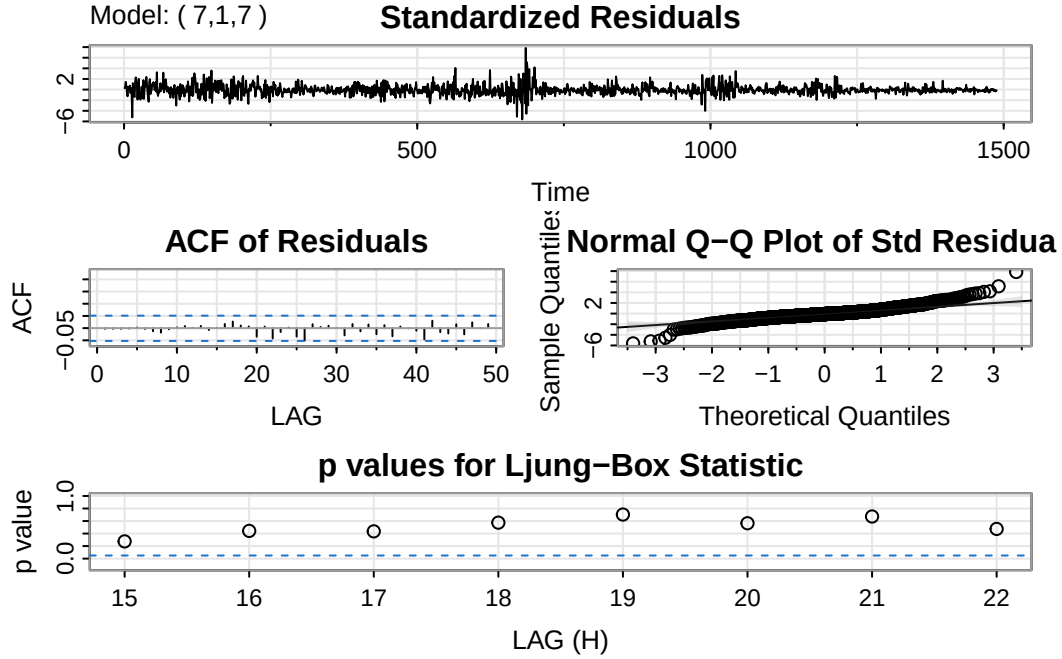
3.1 Fitting a regression model on price

However, the assumptions on regression were not satisfied. The price data is non-stationary, and the QQ plot of the residuals indicates a clear deviation from normality. The Augmented Dickey-Fuller (ADF) test on the model residuals yielded a p-value of 0.0575, failing to reject the null hypothesis of a unit root using 0.01 as the significance level. This suggests that the residuals remain non-stationary.

A robust regression approach was considered too. However, since robust methods do not resolve issues of non-stationarity, this approach was not considered in further analysis. As the residuals are not stationary, an ARMA model cannot be appropriately fitted, and thus no further modelling was performed with regression.

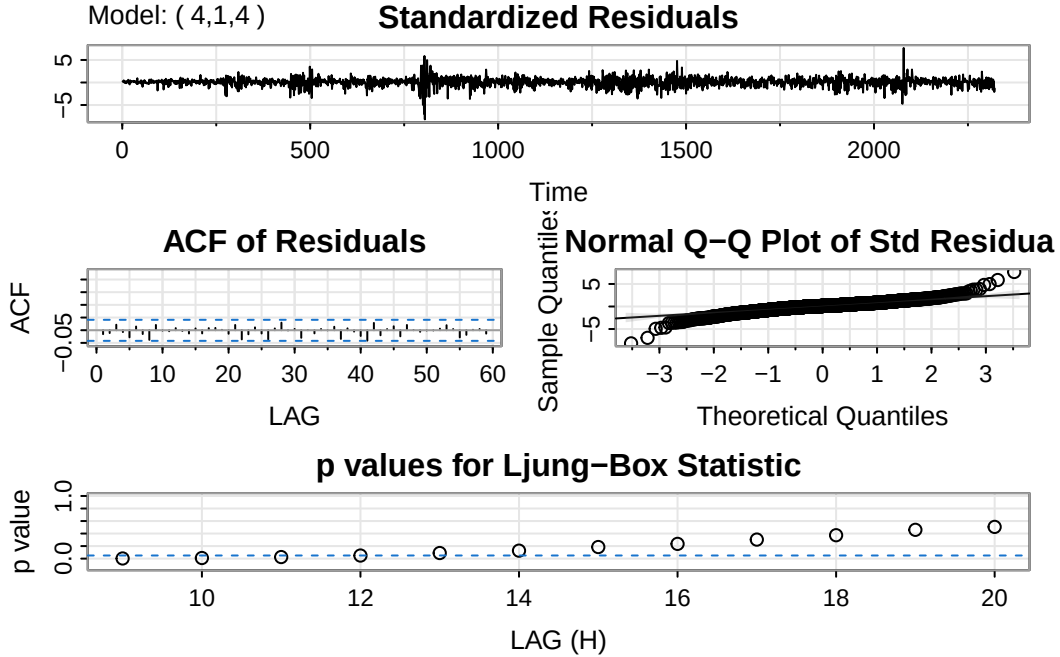


We then apply three “benchmark” forecasting methods: the mean method, the naïve method, and the seasonal naïve method. Mean method forecasts all future values as equal to the average (or mean) of the training set. Naïve method set all forecast values to be the value of the last observation of the historical data. Seasonal naïve method set each forecast to be equal to the last observed value from the same season. In this case, we tested for weekly seasonality using this method. The drift method, which accounts for the amount of change over time, and the decomposition method, which captures the underlying trend, are both considered unnecessary as differencing has already been applied to the log return data.

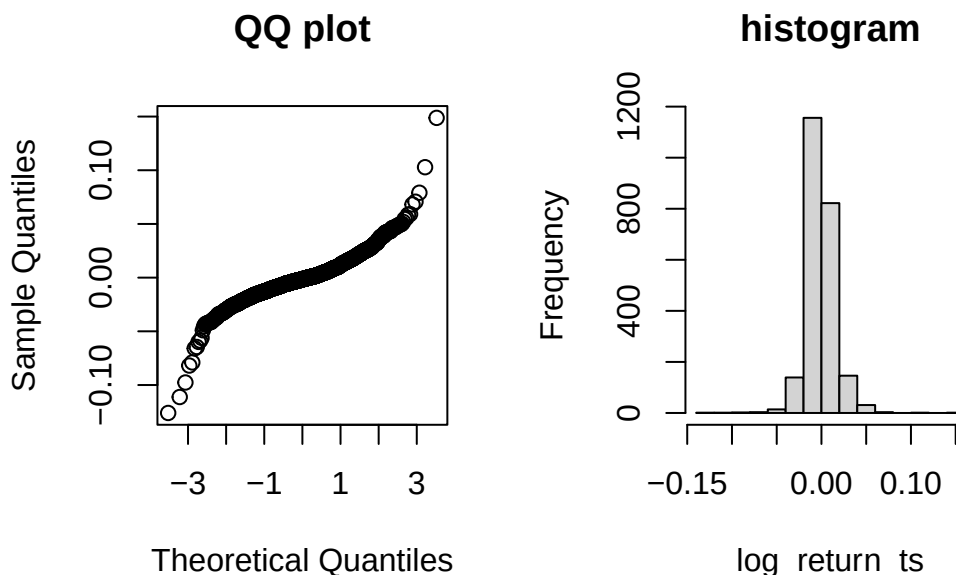


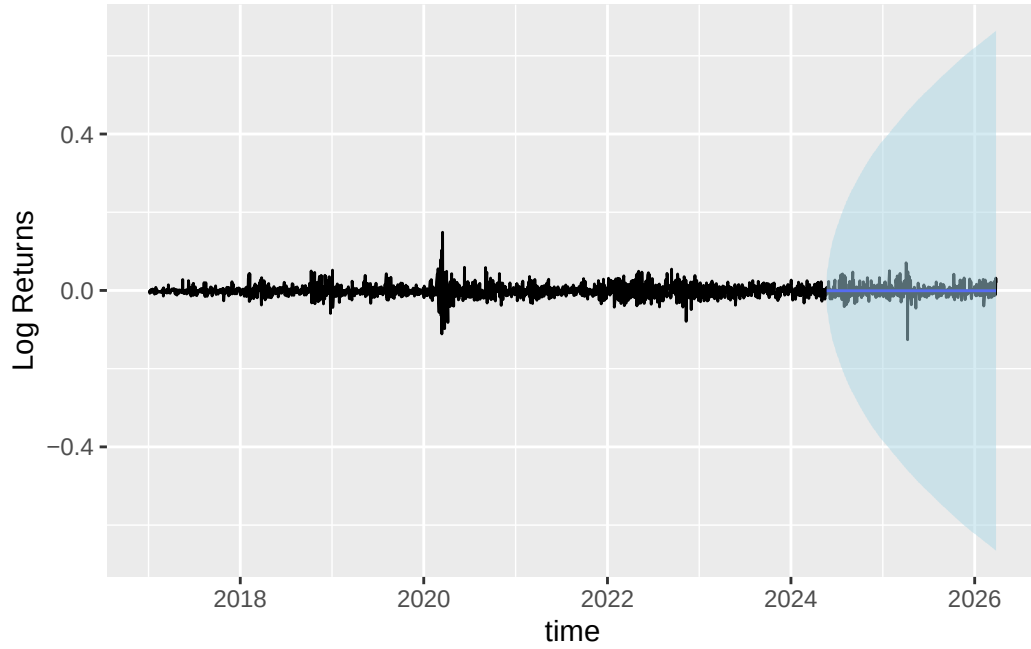
3.4 Full time frame

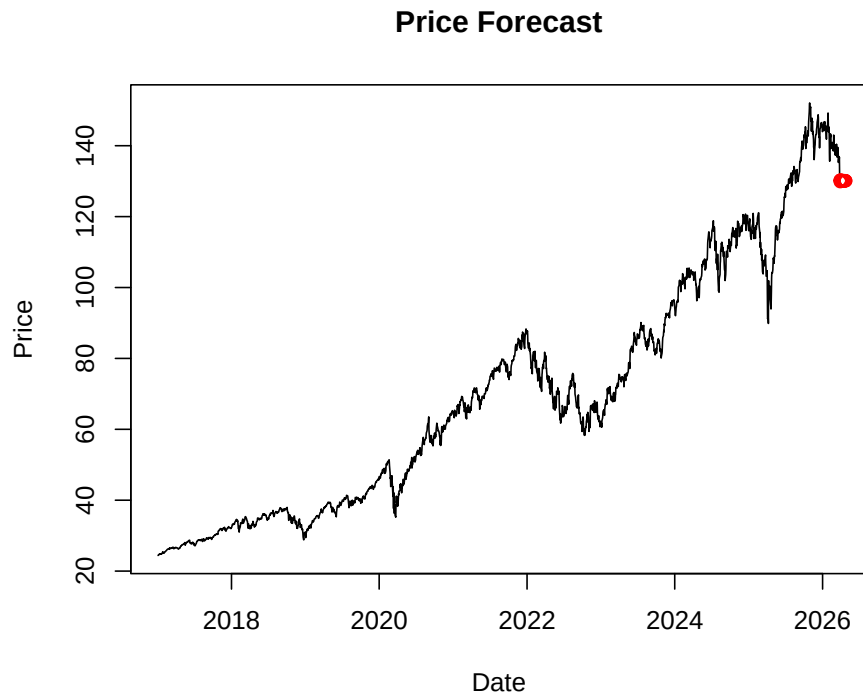
Using the ACF and PACF plots, we can determine the appropriate order of the ARIMA model. According to Figure 5, the ACF gradually tails off, suggesting the presence of autoregressive behavior. According to the PACF, there are strong spikes at lags 2, 4, 7, and 8, and several other significant lags. Based on these patterns, ARIMA(1,1,1), ARIMA(2,1,2), ARIMA(4,1,4), ARIMA(7,1,7), and ARIMA(8,1,8) models were fitted to the price data to capture the time series



To evaluate whether ARIMA(4,1,4) outperforms the mean method, we split the full time frame into 80% training and 20% testing, fitted ARIMA(4,1,4) on the training set, and computed forecast accuracy metrics as shown in `?@tbl-arima_forecast_check`. Comparing with Table 4, the MASE values are the same across both models, however ARIMA(4,1,4) achieves lower MAE, RMSE, and RMSSE, suggesting it is the better performing model.







4 Conclusions

This project tests whether the rise of AI has significantly affected the behavior of technology sector stock using the XLK ETF price data from 2019 to 2026. Using regression, we compared the price over time before and after the appearance of ChatGPT. It suggests that price after AI increased more quickly than before. However, since price data are non stationary, this result should be interpreted with caution. To address this issue, we transform the data into log return to make the data stationary. Based on this transformation, we find no significant evidence that AI has had a meaningful effect on stock returns. An ARIMA(4,1,4) model is then fitted to capture

external variables, excluding these factors may limit the strength and forecasting accuracy of the models.

6 Future Directions

To account for external factors that might affect the stock market, we could add these as predictors in our model. This can help us better estimate stock prices.

Although we have not been able to conclude AI has significant impact on the behavior of technology sector stock returns, it is still useful to estimate how sensitive technology sector stock price comparing